Phase-Aware Adaptive Risk SMPC: Current Progress

1. Problem Formulation

This work studies an unsignalised give-way intersection in a right-hand traffic system using CARLA. The ego vehicle turns left across an oncoming straight-moving priority vehicle, so it must yield, avoid collision, complete the turn, and enter the correct lane.

The main research question is whether an SMPC planner can use interaction-dependent risk allocation to behave more cautiously during safety-critical phases, while avoiding unnecessary conservatism after the priority vehicle has left the conflict zone.

2. Method

The current system combines two components:

- a rule-aware supervisor that enforces right-of-way and final safety;
- an SMPC planner that handles multimodal prediction uncertainty with chance constraints.

The main comparison is:

Policy	Description
`smpc_fixed_risk`	fixed-risk SMPC baseline
`smpc_var_risk`	phase-aware adaptive-variable-risk SMPC

Only `smpc_var_risk` receives adaptive chance-constraint tightening, while the fixed-risk baseline keeps a static risk level.

3. Current Result

The latest main experiment shows that:

Policy	Solver failure max / mean	Footprint separation min / mean (m)	Collision	Yield OK	Completion
`smpc_fixed_risk`	0.0244 / 0.0057	0.9596 / 2.1504	False	True	True
`smpc_var_risk`	0.0293 / 0.0062	0.8745 / 2.1504	False	True	True

Both policies passed all safety gates. The adaptive-risk policy did not introduce collision, yield-rule violation, or task-completion failure, although it has a slightly higher solver cost.

4. Phase-Aware Risk Evidence

The adaptive policy applies stronger risk tightening before the target vehicle clears the conflict zone, then relaxes the tightening after clearance.

Bucket / Phase	Adaptive tightening	Fixed tightening	Adaptive - Fixed
approach / pre-clearance	1.6800	1.6400	+0.0400
critical / pre-clearance	1.7997	1.6400	+0.1597
critical / post-clearance	1.2816	1.6400	-0.3584
near / post-clearance	1.2816	1.6400	-0.3584

This supports the intended mechanism: the SMPC layer is more conservative before target clearance and less conservative after clearance.

5. Ablation Result

An ablation was used to test whether the pre-clearance risk floor is responsible for the stronger adaptive-risk behaviour; when the floor is disabled, the critical pre-clearance tightening gap drops substantially:

Variant	Critical pre-clearance var-fixed tightening gap	Floor applied fraction	Safety gate
phase_floor	+0.1600	1.0000	PASS
no_phase_floor	+0.0603	0.0000	PASS

The gap drops from about +0.160 to +0.060 without the floor. This suggests that the phase-aware risk floor is the key component that makes the adaptive-risk policy clearly more cautious in the critical pre-clearance phase.

6. Next Steps

- Conduct more comprehensive ablation studies on the adaptive-risk design, including the phase floor, risk-tightening levels, post-clearance relaxation, and interaction-severity mapping.
- Further investigate ways to improve the performance of variable-risk SMPC, since the current fixed-risk and variable-risk policies show similar overall safety outcomes. The next goal is to make the adaptive-risk policy provide clearer benefits in safety margin, smoothness, or efficiency while preserving the current safety guarantees.